

Li-S Batteries

How to cite: *Angew. Chem. Int. Ed.* **2022**, 61, e202209350

International Edition: doi.org/10.1002/anie.202209350

German Edition: doi.org/10.1002/ange.202209350

High-Entropy Prussian Blue Analogues and Their Oxide Family as Sulfur Hosts for Lithium-Sulfur Batteries

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Abstract: The introduction of high-entropy into Prussian blue analogues (PBAs) has yet to attract attention in the field of lithium-sulfur battery materials. Herein, we systematically synthesize a library of PBAs from binary to high-entropy by a facile coprecipitation method. The coordination environment in PBAs is explored by X-ray absorption fine structure spectroscopy, which together with elemental mapping confirm the successful introduction of all metals. Importantly, electrochemical tests demonstrate that high-entropy PBA can serve as polysulfide immobilizer to inhibit shuttle effect and as catalyst to promote polysulfides conversion, thereby boosting its outstanding performance. Additionally, a variety of nanocubic metal oxides from binary to ternary are fabricated by using PBAs as sacrificial precursors. We believe that a wide range of new materials obtained from our coprecipitation and pyrolysis methodology can promote further developments in research on PBA systems and sulfur hosts.

Introduction

In recent years, the introduction of high-entropy (HE) into various materials for different applications has aroused rising attention of researchers and facilitated the fast development of a sequence of single-phase multicomponent (equimolar) materials. In disordered multicomponent systems, large configurational entropy is commonly regarded to stabilize the crystal structure, and foster their chemical and structural diversity. Until now, a great quantity of HE materials, represented by initial high-entropy alloys (HEAs)^[1,2] and subsequent high-entropy oxides (HEOs),^[3] have been employed in the fields of environmental protection, electrochemical energy storage, and catalysis. Several recent reports have shown that the battery materials obtained by the introduction of HE can significantly enhance cyclic performance, such as HEOs,^[4,5] high-entropy

oxyfluorides (HEOFs)^[6] and high-entropy MXenes (HEMXs).^[7]

A class of materials which has yet to draw concern in the field of HE lithium-sulfur (Li-S) battery materials are Prussian blue analogues (PBAs). PBAs are a class of perovskite-type materials described by the general formula $A_xM_y[M'(CN)_6]_z$, where A is alkali metal ions such as Na^+ or K^+ , M and M' are di or trivalent transition metal ions.^[8] In such materials, carbon-coordinated M' cations and nitrogen-coordinated M cations are bridged by cyanide groups to form an open framework. In addition, PBAs have the advantages of strong structural stability, simple synthetic procedure, low price, and non-toxic nature, which promote their wide application in the field of sensors, electrocatalysis, photocatalysis, gas storage, and sodium-ion batteries. Nevertheless, in the past, few studies were performed on the utilization of PBAs as sulfur hosts in Li-S batteries. As a matter of fact, as host materials of Li-S batteries, PBAs have the following advantages. First, as the Lewis acid sites, the open metal centers have good affinity with negatively charged polysulfide anions, thus preventing the shuttling of lithium polysulfides (LiPS).^[9,10] Second, their open framework structures can immobilize LiPS via the pore confining effect and alleviate volumetric change in the process of battery charge/discharge.^[11] Third, their geometry changes little in the process of ion insertion because of stable structure, which further alleviates the issue of capacity attenuation.^[12] Fourth, these compounds are fabricated via a facile and inexpensive coprecipitation approach using water as solvent. Finally, they are highly tunable, and the formed HE-PBA provides more possibilities for the realization of excellent Li-S battery cathode materials. Thus, further research on HE-PBAs as sulfur hosts is imperative.

Herein, based on five metal cations (Mn^{2+} , Co^{2+} , Ni^{2+} , Cu^{2+} , and Zn^{2+}) and $K_3[Fe(CN)_6]$, we report the successful synthesis of 16 kinds of PBAs from binary to high-entropy by a facile coprecipitation method with water as solvent. Among these PBAs, we choose four kinds of PBAs to explore the influence of the introduction of different metals on the coordination environment based on X-ray absorption fine structure (XAFS) spectroscopy, which together with elemental mapping analysis confirm the successful introduction of all metals. In addition, NiFe-, NiCuFe-, CoNiCuFe-, and CoNiCuMnZnFe-PBA (HE-PBA) nanocubes are employed as sulfur host materials for Li-S batteries. LiPS adsorption, in situ ultra-violet visible (UV/Vis), and cyclic voltammetry (CV) measurements reveal that HE-PBA can not only serve as a polysulfide immobilizer to enhance the

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suppression of the polysulfide shuttle effect, but also as catalyst to promote polysulfides conversion. Electrochemical tests show that HE-PBA-S cathode achieves the outstanding performance with a capacity of 570.9 mAh g^{-1} over 200 cycles at 0.1 C. Besides, PBAs are used as sacrificial precursors to control the synthesis of a wide range of metal oxides with nanocubic structure from binary to senary, which provide practical guidance for the synthesis of other PBA derivatives. We believe that a wide range of new materials obtained from our coprecipitation and pyrolysis methodology can promote further developments in research on PBA systems and provide a valuable reference for the selection of sulfur host materials.

Results and Discussion

Five metal cations— Mn^{II} , Co^{II} , Ni^{II} , Cu^{II} , and Zn^{II} —from metal nitrates, are selected as metal sources to fabricate a series of binary, ternary, quaternary, and HE PBAs using water as solvent by a simple room temperature coprecipitation method (Figure 1). These PBAs are named according to the combination of metal elements. In the synthesis process, the amount of $\text{K}_3[\text{Fe}(\text{CN})_6]$ is fixed at 2 mmol, and the total moles of added other metal salts remain unchanged at 2 mmol. The detailed steps of the synthetic process can be found in the Supporting Information.

The morphologies were characterized through scanning electron microscopy (SEM) and transmission electron microscopy (TEM) (Figure 2). For binary PBAs, NiFe-PBA,

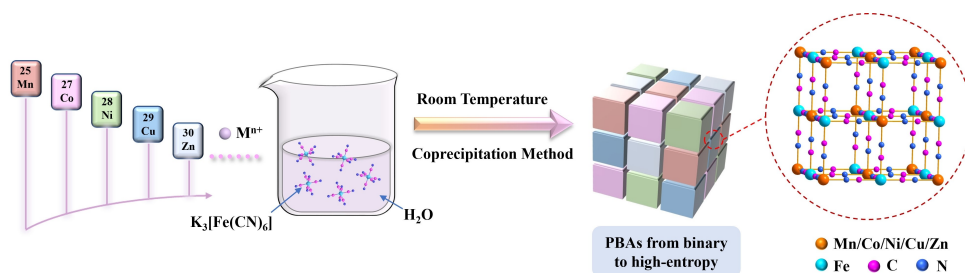


Figure 1. Schematic illustration of the synthetic process for PBA family from binary to high-entropy system by coprecipitation method at room temperature.

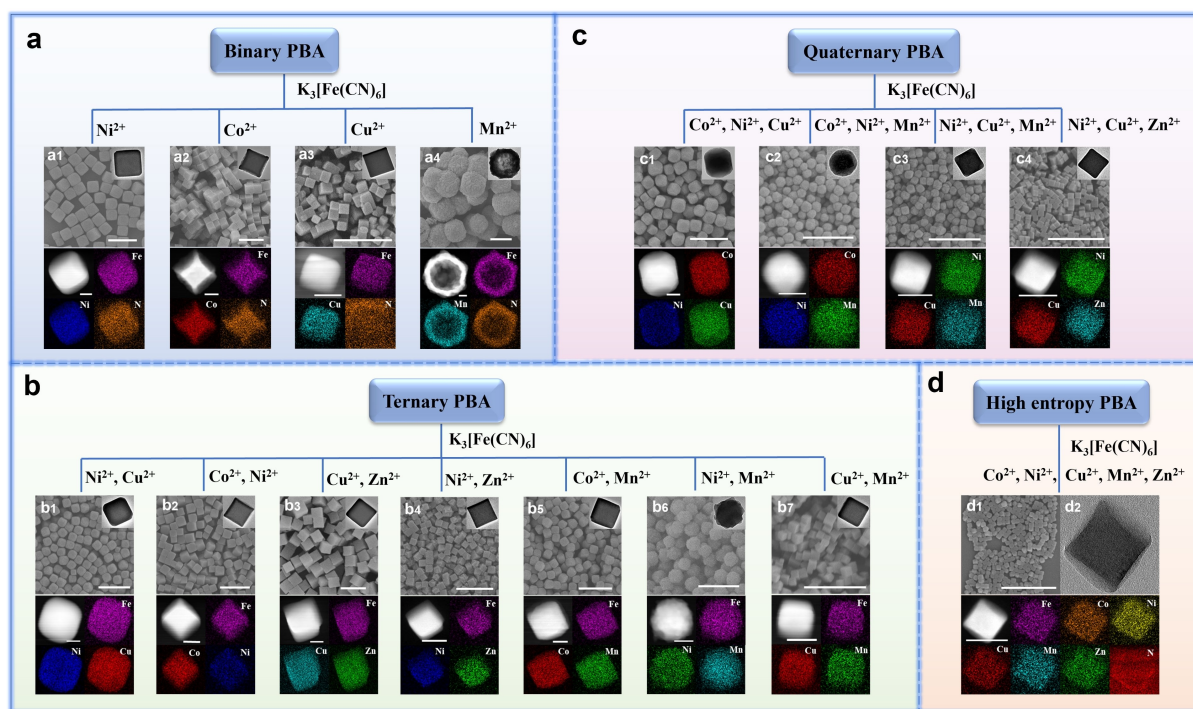


Figure 2. Morphology and composition characterizations of sixteen kinds of PBAs. SEM images, TEM images and the corresponding elemental mapping analysis of binary PBAs (a1–a4; representing NiFe-PBA, CoFe-PBA, CuFe-PBA, and MnFe-PBA in turn), ternary PBAs (b1–b7; representing NiCuFe-PBA, CoNiFe-PBA, CuZnFe-PBA, NiZnFe-PBA, CoMnFe-PBA, NiMnFe-PBA, and CuMnFe-PBA in turn), quaternary PBAs (c1–c4; representing CoNiCuFe-PBA, CoNiMnFe-PBA, NiCuMnFe-PBA, and NiCuZnFe-PBA in turn), and HE-PBA (d1 and d2; representing CoNiCuMnZnFe-PBA). The scale bars for SEM and HAADF-STEM images indicate 1 μm and 100 nm, respectively.

CoFe-PBA, and CuFe-PBA exhibit regular cubic shape with particle sizes of ≈ 310 , ≈ 300 , and ≈ 120 nm, respectively (Figure 2a1–a3). Distinctively, MnFe-PBA shows a uniform hollow sphere shape with rough surface, and the particle size is about 1 μm (Figures 2a4 and S1). However, ZnFe-PBA does not form a precipitate by this approach. The formation of ternary PBAs is controlled by adjusting the molar ratio of Mn, Co, Ni, Cu, Zn. By controlling 1:3 molar ratio of Cu/Ni, Ni/Co, Cu/Zn, Zn/Ni, Mn/Co, Mn/Cu, and Ni/Mn, the nanocubic NiCuFe-PBA (≈ 300 nm), CoNiFe-PBA (≈ 200 nm), CuZnFe-PBA (≈ 420 nm), NiZnFe-PBA (≈ 140 nm), CoMnFe-PBA (≈ 260 nm), CuMnFe-PBA (≈ 100 nm), and spherical NiMnFe-PBA (≈ 290 nm) were successfully prepared (Figure 2b). In addition, the CoNiFe-PBA-2, NiCuFe-PBA-2, CuZnFe-PBA-2, NiZnFe-PBA-2, CoMnFe-PBA-2, NiMnFe-PBA-2, and CuMnFe-PBA-2 were synthesized by 3:1 molar ratio of Ni/Co, Cu/Ni, Cu/Zn, Zn/Ni, Mn/Co, Ni/Mn, and Mn/Cu (Figure S2). Interestingly, the morphology of CoMnFe-PBA-2 changed from the original cubic to spherical shape. For quaternary PBAs, cubic CoNiCuFe-PBA (≈ 330 nm), NiCuMnFe-PBA (≈ 110 nm), NiCuZnFe-PBA (≈ 90 nm), and spherical CoNiMnFe-PBA (≈ 150 nm) were prepared in 2:1:1 molar ratio of Co/Ni/Cu, Ni/Cu/Mn, Ni/Cu/Zn, and Co/Ni/Mn, respectively (Figure 2c). More importantly, uniform cubic HE-PBA with a particle size of ≈ 80 nm was also fabricated in 1:1:1:1 molar ratio of Co/Ni/Cu/Mn/Zn (Figure 2d). The high-angle annular dark field scanning TEM (HAADF-STEM) images and corresponding element mapping analysis demonstrate the homogeneous distribution of Mn, Fe, Co, Ni, Cu, and Zn in PBA face-centered cubic framework. Based on the above analysis, it is found that the synthesized spherical PBAs have a common feature, that is, they all contain Mn source. It can be concluded that for PBAs which synthesized by coprecipitation method with water as solvent and without Cu source, the introduction of Mn source or the increase of its content tends to synthesize spherical PBAs.

Powder X-ray diffraction (PXRD) was conducted to study the crystal structure and phase purity of PBA family samples. The PXRD patterns of all PBAs exhibit a set of characteristic diffraction peaks of PBA, and no superfluous peaks for other phases are observed, indicating a single-phase PBA structure (Figures S3–S6).^[13] The enlarged view of XRD patterns of NiFe-PBA, NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA display that the (200) peaks of PBA are at 17.3° , 17.7° , 17.5° , 17.6° , respectively (Figures S7). This implies that compared with binary PBAs, the introduction of multi-metals would cause a certain degree of lattice shrinking.^[7] In addition, the composition of Fe, Co, Ni, Cu, Mn and Zn were mensurated by inductively coupled plasma-atomic emission spectrometry (ICP-AES), as shown in Table S1, demonstrating the successful formation of this series of PBAs. XAFS measurements, a puissant technique to investigate the local coordination environment, were performed to explore the influence of the introduction of Fe, Co, Ni, Cu, Mn and Zn in PBA framework on the structure and judge their valence state. We consider the following four kinds of PBAs as example. X-ray absorption near edge structure (XANES) spectra at the Fe K-edge for

NiFe-PBA, NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA are shown in Figure 3a, with Fe foil and Fe_2O_3 as references. It can be observed that the curve trend of these PBAs are significantly different from that of the Fe foil but similar to that of Fe_2O_3 , indicating that the Fe has a similar oxidation state with that in Fe_2O_3 .^[14] The absorption edge position of Fe has weakly shifted to low energy level according to the order of NiFe-PBA, NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA, which further suggests that the doping of different metals lead to the change of the electronic property.^[15] The XANES profiles of all Fe are nearly identical, strongly suggesting that the same coordination environment is adopted by Fe moieties in different PBAs. Based on the XANES of Co, Ni, and Cu K-edge in different PBAs, the same conclusion can be drawn. XANES spectra for these PBAs at the Co, Ni, Cu, Mn and Zn K-edge are similar to those of CoO, NiO, CuO, MnO and ZnO (Figure 3c, e, g, i, and k), respectively, indicating that the stable valence states for these five metal elements are about +2. The Fe, Co, Ni, Cu, Mn and Zn K-edge extended X-ray absorption fine structure (EXAFS) spectrum for these PBAs reveals obvious Fe–C, Co–N, Ni–N, Cu–N, Mn–N and Zn–N (Figure 3b, d, f, h, j and l),^[16–19] further suggesting that Fe, Co, Ni, Cu, Mn, and Zn are successfully introduced into the PBA framework. Among these chemical bonds, the introduction of different metals has a greater impact on the Fe–C bond length, but a less impact on the bond length of Co–N, Ni–N, and Cu–N.

In order to further expand PBA materials, we also explored the controllable synthesis of PBA derivatives. A series of metal oxides (Figure 4a), including NiFe-oxide, CoFe-oxide, CoNiFe-oxide, NiCuFe-oxide, NiZnFe-oxide, CuZnFe-oxide, CoMnFe-oxide, and CoNiCuMnZnFe-oxide, were fabricated by the pyrolysis at 500°C in air for 2 h of NiFe-PBA, CoFe-PBA, CoNiFe-PBA, NiCuFe-PBA, NiZnFe-PBA, CuZnFe-PBA, CoMnFe-PBA, and HE-PBA, respectively. The corresponding PXRD patterns of these metal oxides are presented in Figure S8. SEM images display these oxides retain their original cubic morphology, and TEM images demonstrate the hollow architecture of NiCuFe-oxide, NiZnFe-oxide, and CoMnFe-oxide (Figures 4b–i and S9). The high-resolution TEM (HRTEM) images exhibit obvious lattice fringe, which together with the selected area electron diffraction (SAED) pattern confirm the good crystalline characteristics of CoNiCuMnZnFe-oxide (Figure 4j, k). The elemental mapping images for CoNiCuMnZnFe-oxide illustrate that Fe, Co, Ni, Cu, Mn, Zn, and O elements are homogeneously distributed in the porous nanocube (Figure 4l). The X-ray photoelectron spectroscopy (XPS) was utilized to explore the surface chemical compositions and the valence of elements of CoNiCuMnZnFe-oxide (Figure S10). The high-resolution XPS spectra show that Cu and Zn cations mainly exist in the form of Cu^{2+} and Zn^{2+} , respectively.^[20] The Fe 2p, Co 2p, Ni 2p, and Mn 2p spectra indicate that there are corresponding divalent and trivalent species.^[21–23] Notably, the O 1s spectra at 529.5 eV corresponds to oxygen atoms bound to metals.^[24] The chemical structure of PBAs and oxides was further investigated by Fourier transform infrared (FTIR) spectro-

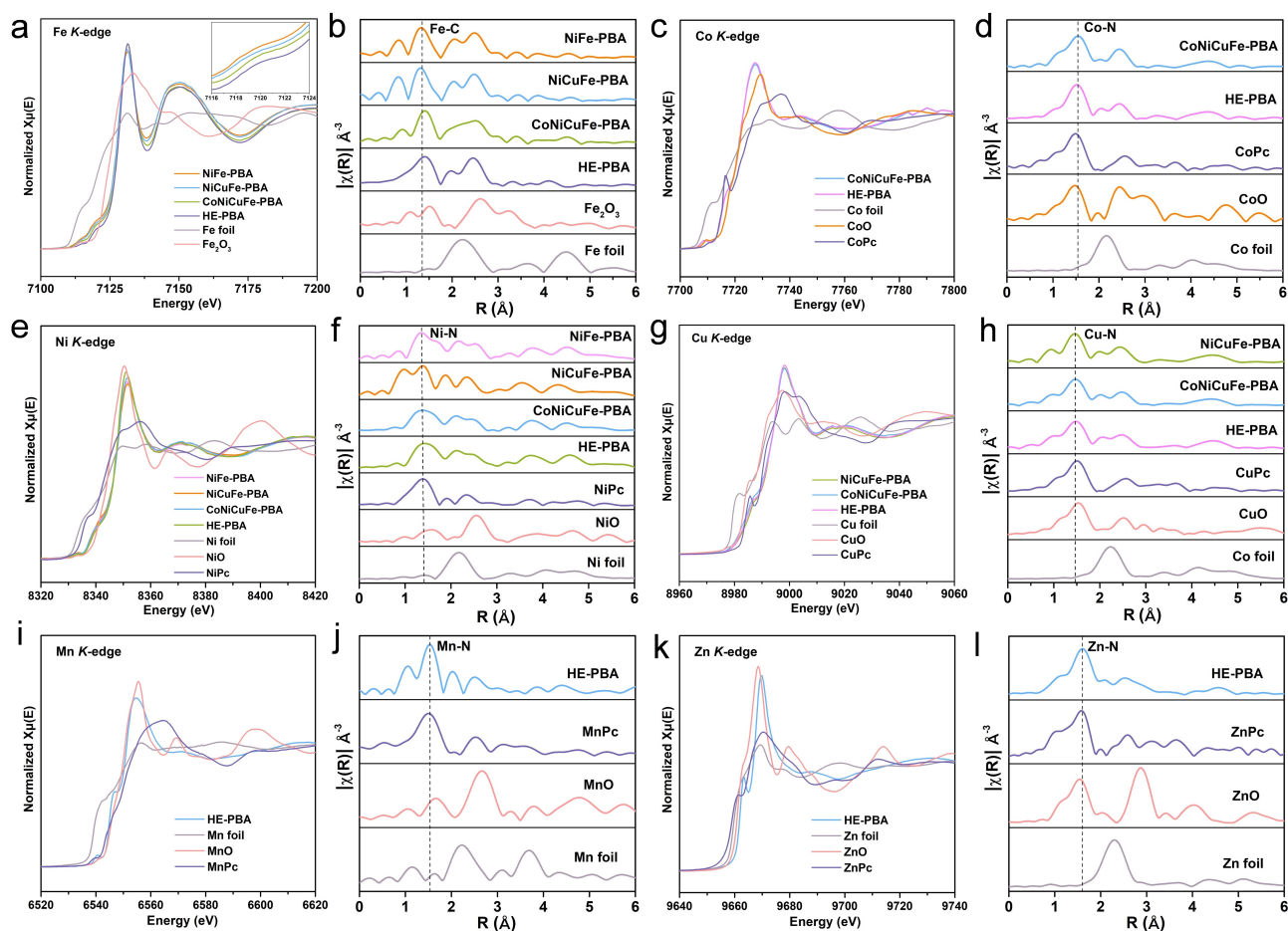


Figure 3. Chemical state and atomic coordination environment of seven kinds of PBAs. a) Fe K-edge XANES spectra with Fe foil, and Fe_2O_3 as references, and corresponding b) EXAFS spectra for NiFe-PBA, NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA. The inset shows the enlarged view of pre-edge portion. c) Co K-edge XANES spectra with Co foil, CoO, and Co-coordinated phthalocyanine (CoPc) as references, and corresponding d) EXAFS spectra for CoNiCuFe-PBA and HE-PBA. e) Ni K-edge XANES spectra with Ni foil, NiO, and NiPc as references, and corresponding f) EXAFS spectra for NiFe-PBA, NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA. g) Cu K-edge XANES spectra with Cu foil, CuO, and CuPc as references, and corresponding h) EXAFS spectra for NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA. i) Mn K-edge XANES spectra with Mn foil, MnO, and MnPc as references, and corresponding j) EXAFS spectra for HE-PBA. k) Zn K-edge XANES spectra with Zn foil, ZnO, and ZnPc as references, and corresponding l) EXAFS spectra for HE-PBA.

scopy analysis. The weak peak at 1607 cm^{-1} and the broad peak at 3403 cm^{-1} in PBAs originate from the $-\text{OH}$ bending vibration and stretching vibration of H_2O , respectively (Figure S11).^[25] The typical $\text{C}\equiv\text{N}$ group stretching vibration can be observed from the adsorption bands in the range of $2000\text{--}2200\text{ cm}^{-1}$.^[26] After oxidation of the PBA precursors at 500°C , the characteristic peaks of CN disappear (Figure S12), indicating that the PBA precursors are completely decomposed and transformed to metallic oxides. In addition, we measured the work function of NiFe-oxide, CoNiFe-oxide and CoNiCuMnZnFe-oxide using ultraviolet photoelectron spectroscopy (UPS) spectra. Results reveal that the work function of CoNiCuMnZnFe-oxide (4.53 eV) is lower than that of NiFe-oxide (4.78 eV) and CoNiFe-oxide (4.70 eV) (Figure S13), indicating that the CoNiCuMnZnFe-oxide possesses better electronic properties,^[27] which provides a valuable reference for the research of multi-metal oxides. Interestingly, several metal phosphides (NiFe-phosphide, CoFe-phosphide, CoNiFe-phosphide, and

CoMnFe-phosphide) were prepared by thermal treatment of a series of PBAs. The corresponding XRD patterns, SEM, and TEM images are presented in Figure S14. It can be observed that CoFe-phosphide exhibits a cubic morphology with a hollow structure, and CoNiFe-phosphide displays a slightly truncated construction on the eight corners of the cube.

Through melt-diffusion approach at 130°C , sulfur molecules were well introduced into the pores of NiFe-PBA, NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA, labeled as NiFe-PBA-S, NiCuFe-PBA-S, CoNiCuFe-PBA-S, and HE-PBA-S. The PXRD patterns of PBA-S samples indicate that these PBA samples are stable in the process of sulfur melt and impregnation, and do not decompose at 130°C (Figure S15). To illustrate the interaction between HE-PBA and sulfur, the XPS characterization of HE-PBA-S and pristine sulfur was executed. A broad peak located at 168.8 eV suggests the formation of sulfate species. Moreover, it can be clearly observed that the S 2p spectra in HE-PBA-S has

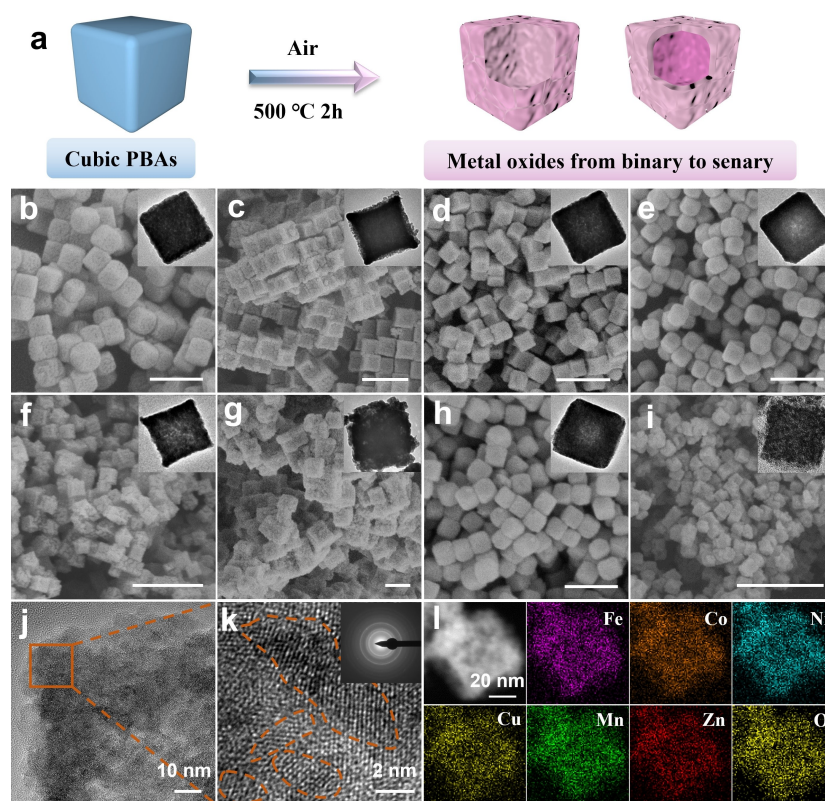


Figure 4. Morphology characterization of metal oxides. a) The synthesis procedure for metal oxides. SEM and TEM images of b) NiFe-oxide, c) CoNiFe-oxide, d) NiCuFe-oxide, e) NiZnFe-oxide, f) CuZnFe-oxide, g) CoMnFe-oxide, h) CoNiCuMnZnFe-oxide, and i) CoNiCuMnZnFe-oxide. The scale bars for SEM images indicate 500 nm. j, k) HRTEM images and SAED pattern of CoNiCuMnZnFe-oxide. l) HAADF-STEM image and corresponding elemental mapping of CoNiCuMnZnFe-oxide.

an obvious chemical shift compared with that of pristine sulfur (Figure S16), that is from original 165.2 eV to 164.6 eV of S 2p_{1/2} and 164.0 eV to 163.4 eV of S 2p_{3/2}. This confirms the great interaction between sulfur and HE-PBA, and the applicability of HE-PBA as sulfur carriers, which can be fixed in the cathode to suppress the shuttle effect and heighten the cyclic performance for Li–S batteries.^[28] By governing the feeding ratio, the sulfur content was controlled precisely, and the accurate amount of sulfur loading was quantitatively analyzed by both thermal-gravimetric analysis (TGA) (Figure S17) and elemental analysis (Table S2). Given the high porosity of PBAs, sulfur can be introduced at approximately 70 % by weight. Additionally, we evaluated the influence of sulfur infiltrated into the four sulfur hosts by N₂ adsorption isotherms. It can be found that the N₂ absorption and pore volume of four sulfur-loaded samples are significantly reduced compared with the pristine PBAs (Figures S18–S25), indicating that sulfur is magnificently embedded into these PBA pores.^[29] The HE-PBA displays the highest N₂ uptake capacity, suggesting that it possesses enough free space for sulfur injection. Moreover, for these PBA host materials, the pore-size distribution curves, measured by the Barrett–Joyner–Halenda approach, display a broad pore distribution (1.7–35 nm), which contribute to immobilizing sulfur and polysulfides in the porous framework of PBAs.

The influence of binary, ternary, quaternary, and HE PBAs on the electrochemical performance was studied. To confirm the redox reaction between sulfur and LiPS, CV tests of NiFe-PBA-S, NiCuFe-PBA-S, CoNiCuFe-PBA-S, and HE-PBA-S were performed in the range of 1.7–2.7 V at a scan rate of 0.1 mVs^{−1} (Figure 5a). As far as HE-PBA-S cathode is concerned, there are two apparent cathodic peaks at 2.05 and 2.30 V, which can be attributed to the transformation from S₈ molecule to high-order LiPS and further reduction to Li₂S₂ and Li₂S, respectively.^[30] The anodic peak at 2.41 V can be ascribed to the reverse oxidation process from solid Li₂S to LiPS and ultimately to elemental sulfur. The CV curve of HE-PBA-S cathode shows obvious positive shift in reduction peak and negative shift in oxidation peak compared with the other three electrodes, which indicates that the polarization degree is reduced and the reaction kinetics is rapid.^[31] The electrochemical impedance spectroscopy (EIS) was employed to further estimate the kinetic behaviors of the NiFe-PBA-S, NiCuFe-PBA-S, CoNiCuFe-PBA-S, and HE-PBA-S electrodes (Figure 5i). The Nyquist plots of these cathodes consist of a semicircle in the high-frequency region and a sloping line in the low-frequency region, representing the charge transfer resistance (*R*_{ct}) and Warburg impedance (*W*_o), respectively. According to the equivalent circuit, the fitted kinetic parameters are presented in Table S3. It can be seen clearly that HE-PBA-S

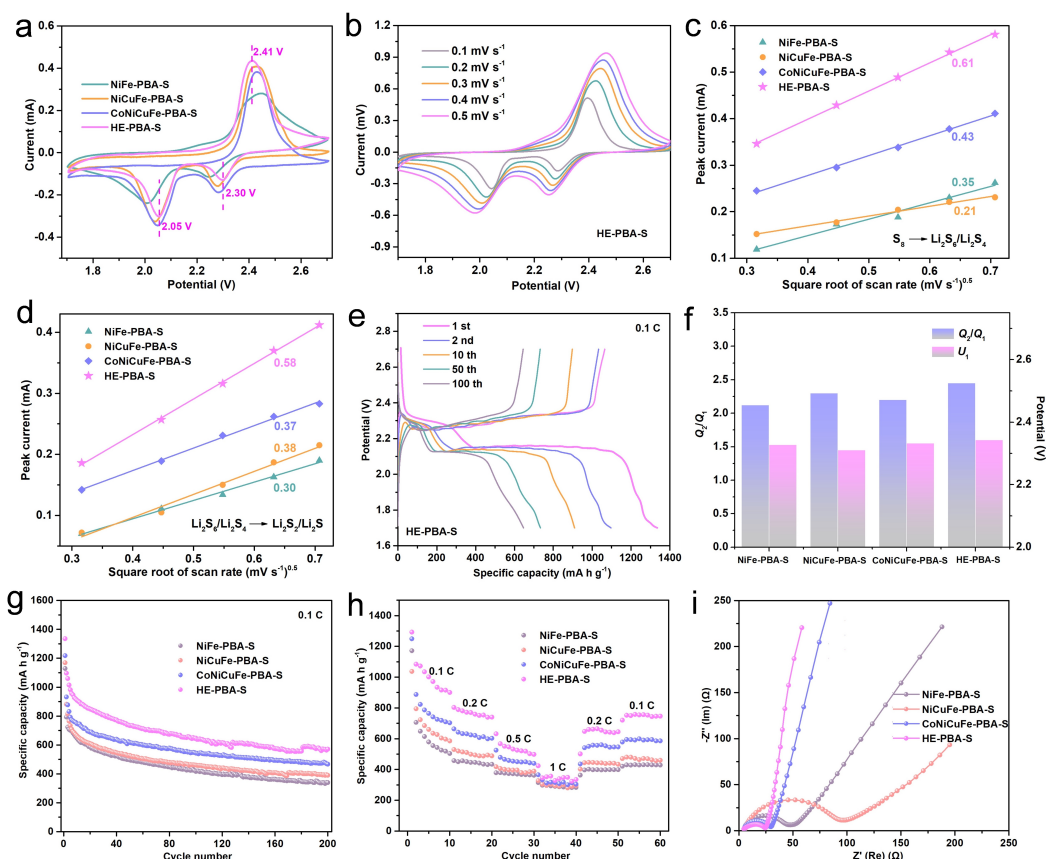


Figure 5. Li-S battery performance evaluation. a) CV curves of the first cycle of NiFe-PBA-S, NiCuFe-PBA-S, CoNiCuFe-PBA-S, and HE-PBA-S cathode at 0.1 mV s^{-1} . b) CV curves at different scan rates of the HE-PBA-S cathode. The plots of peak current versus the square root of the scan rate ($v^{0.5}$) for c) the conversion process of S_8 into $\text{Li}_2\text{S}_6/\text{Li}_2\text{S}_4$ and d) the conversion process of $\text{Li}_2\text{S}_6/\text{Li}_2\text{S}_4$ into $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$. e) The GCD profiles at 0.1 C of HE-PBA-S electrode. f) Value of Q_2/Q_1 and U_1 (collected from GCD profiles at 0.1 C). g) Cycle performance at 0.1 C over 200 cycles. h) Rate performance of at different current densities. i) EIS spectra for four PBA-S samples.

electrode has the smallest R_{ct} (21.85Ω), indicating enhanced charge transfer at the electrode/electrolyte interface. The steeper slope of the Warburg tail for the HE-PBA-S electrode confirms the faster Li^+ diffusion process. In addition, two parameters, U_1 and Q_2/Q_1 , are introduced to represent onset potential and discharge capacity ratio, respectively. Among the four PBA-S electrodes, HE-PBA-S achieves the highest U_1 of 2.35 V and the highest Q_2/Q_1 of 2.45 at 0.1 C (Figure 5f), indicating an accelerated Li^+/e^- transport within the HE-PBA host.^[32] All of these results demonstrate that the HE-PBA can promote efficient charge transfer and LiPS redox reaction of sulfur cathode in Li-S batteries.

To confirm the catalytic ability of HE-PBA for the conversion of soluble polysulfides, CV tests at various scan rates from 0.1 to 0.5 mV s^{-1} were carried out (Figures 5b and S26). With the augmentation of the scan rate, the shape of redox peaks does not modify, suggesting good electrochemical stability. Moreover, the redox peak currents show a linear relationship with the square root of the scan rate. The diffusion coefficient of Li^+ can be estimated by the Randles-Sevcik equation:

$$I_p = (2.69 \times 10^5) n^{1.5} A D_{\text{Li}}^{0.5} C_{\text{Li}} v^{0.5}$$

where I_p represents the peak current, n is the electron transferred number, A is the surface area of the electrode, D_{Li} is the Li^+ diffusion coefficient, C_{Li} is the Li^+ concentration in the electrolyte, and v is the scan rate. Generally speaking, in the assembled battery, n , A , and C_{Li} can be regarded as constant. Thus, the slope of the plot ($I_p/v^{0.5}$) is often employed to judge the diffusion rate of Li^+ . The fitted line plot in Figure 5c represents the reduction conversion of S_8 into $\text{Li}_2\text{S}_6/\text{Li}_2\text{S}_4$, and the plot in Figure 5d reflects the reduction process of $\text{Li}_2\text{S}_6/\text{Li}_2\text{S}_4$ into $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$. It is found that the slope of HE-PBA-S electrode is higher than those of the NiFe-PBA-S, NiCuFe-PBA-S, and CoNiCuFe-PBA-S electrode, which verifies the faster Li^+ diffusion rate and more efficient polysulfides conversion in HE-PBA-S electrode.^[33–35]

The galvanostatic charge/discharge (GCD) curves at distinct cycles under 0.1 C (Figures 5e and S27) clearly reveal the change in specific capacity and manifest two discharge plateaus corresponding to the characteristic transformation reactions of $\text{S} \rightarrow \text{Li}_2\text{S}_x \rightarrow \text{Li}_2\text{S}_2/\text{Li}_2\text{S}$, which agrees with CV test results. The HE-PBA-S cathode delivers the

highest initial discharge specific capacity ($1335.6 \text{ mAh g}^{-1}$) in comparison with the NiFe-PBA-S ($1129.0 \text{ mAh g}^{-1}$), NiCuFe-PBA-S ($1169.1 \text{ mAh g}^{-1}$), and CoNiCuFe-PBA-S ($1218.3 \text{ mAh g}^{-1}$). The cycling performance of the four PBAs sulfur-loaded samples at a current rate of 0.1 C further verifies the outstanding performance of HE-PBA-S cathode (Figure 5g). It can still maintain an excellent capacity of 570.9 mAh g^{-1} over 200 circulations, far superior to that of the NiFe-PBA-S (339.5 mAh g^{-1}), NiCuFe-PBA-S (389.8 mAh g^{-1}), and CoNiCuFe-PBA-S (467.0 mAh g^{-1}). Moreover, it also achieves a superior coulombic efficiency ($>99\%$ from the eighth circulation) (Figure S28). Even under a higher current density of 0.5 C, the HE-PBA-S cathode delivers a high specific capacity of 424.9 mAh g^{-1} after 200 circulations (Figure S29). Compared with earlier reported test results for metal organic framework-based and PBA-based sulfur hosts (Table S4), HE-PBA-S cathode displays competitive electrochemical performance. In addition, the rate capabilities were measured under a current density range from 0.1 C to 1 C (Figure 5h). It can be found clearly that the HE-PBA-S exhibits a good rate capability with reversible capacities of 1292.6, 803.5, 631.8, and 424.0 mAh g^{-1} at current densities of 0.1, 0.2, 0.5, and 1 C, respectively, indicating its high sulfur utilization rate. More importantly, when the current density is switched back to 0.2 and 0.1 C, HE-PBA-S electrodes can still output a high discharge capacity of 647.4 and 752.6 mAh g^{-1} , respectively, indicating the good stability at various rates.

In situ UV/Vis analysis was used to directly demonstrate the species change of LiPS. The configuration of Li-S cuvette cells enables simultaneous measurements of the UV/Vis spectra of electrolyte and the galvanostatic discharge process. Spectra were measured in the range between 420 and 680 nm during battery operation. Figure 6a and b show the measured UV/Vis spectra from the Li-S batteries using

NiFe-PBA-S and HE-PBA-S as cathode materials, respectively. The periodic UV/Vis spectra were collected during the first discharge process at 0.05 C. Owing to the formation of soluble LiPS, various LiPS with long or short chains can be qualitatively determined. Compared with the NiFe-PBA-S electrode, the HE-PBA-S electrode maintains a longer discharge time. Furthermore, the spectrum of HE-PBA-S electrode shows a higher concentration of S_6^{2-} and S_8^{2-} in comparison with NiFe-PBA-S electrode at the same time, which is ascribed to the faster redox reaction between HE-PBA and LiPS. As the discharge depth enters the second platform, the long-chain LiPS all presents a certain degree of decline. The in situ XRD patterns during the first GCD process at 0.1 C of the NiFe-PBA-S and HE-PBA-S are presented in Figure 6c and d, where the peak at 18.6° and the broad peak around 22.5° are attributed to the Al foil. The peaks located at 14.3° and 17.1° are attributed to the PBA plane. In addition to the characteristic peaks, no significant diffraction signal is observed, which is attributed to the lower content of sulfur species.

In order to further evaluate the adsorption ability of the binary, ternary, quaternary, and HE PBAs with LiPS, we performed the visual LiPS adsorption test of NiFe-PBA, NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA. Result demonstrates the LiPS adsorption effect is consistent with the Li-S battery performance. The absorption peak located at $\approx 420 \text{ nm}$ can be ascribed to Li_2S_4 specie (Figure S30).^[36] Clearly, HE-PBA achieves the fastest color fading. The Li_2S_4 solution adsorbed by the HE-PBA turns transparent after 4 h, which verifies the strong LiPS adsorption ability of the HE-PBA and shows that it can effectively inhibit the shuttle of polysulfides during battery cycling. In general, the HE-PBA as a sulfur host for Li-S battery delivers great potential, which can be attributed to its splendid LiPS

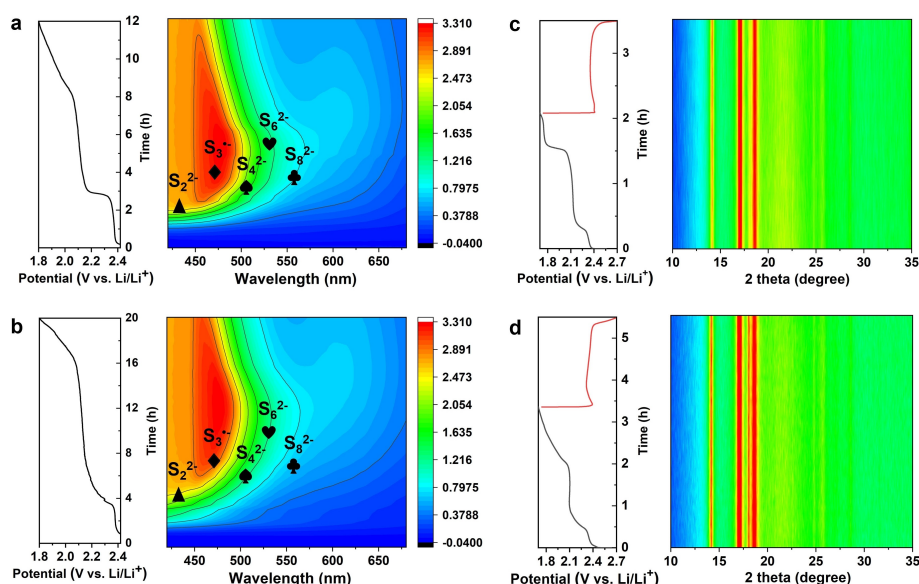


Figure 6. Contour maps of in situ UV/Vis spectra and the corresponding discharge profile of a) NiFe-PBA-S electrode and b) HE-PBA-S electrode. Contour maps of in situ XRD patterns and the corresponding GCD profile of c) NiFe-PBA-S electrode and d) HE-PBA-S electrode.

confinement and catalytic ability, as well as better conductivity.

Conclusion

In summary, we realize room temperature synthesis of a library of PBAs from binary to high-entropy by a facile coprecipitation method. XAFS spectra, elemental mapping, and ICP-AES analysis confirm the successful formation of this series of PBAs. When used as sulfur hosts, it is found that HE-PBA can act as polysulfide immobilizers and catalysts to achieve high-performance Li–S batteries. In addition, a wide range of metal oxides with nanocubic structure from binary to senary are synthesized by using PBAs as sacrificial precursors. UPS spectra analysis shows the CoNiCuMnZnFe-oxide possess good electronic properties, which can prompt the further research of HE-PBA derivatives. This work provides a practical guidance for the selection of HE-PBAs as sulfur host materials and the synthesis of other PBAs and PBA derivatives. Additionally, the results of this study show that the construction of HE materials is an effective strategy for achieving high-performance Li–S batteries, which offer a new direction to construct multifunctional sulfur hosts that meet comprehensive selection criteria with strong binding, high conductivity, and fast kinetics.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (U1904215), Natural Science Foundation of Jiangsu Province (BK20200044), Changjiang scholars program of the Ministry of Education (Q2018270), and Postgraduate Research & Practice Innovation Program of Jiangsu Province (KYCX22_3460). The authors acknowledge the Priority Academic Program Development of Jiangsu Higher Education Institutions.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords: Catalyst • High-Entropy • Lithium-Sulfur Batteries • Polysulfide Immobilizer • Prussian Blue Analogues

- [1] H. Li, Y. Han, H. Zhao, W. Qi, D. Zhang, Y. Yu, W. Cai, S. Li, J. Lai, B. Huang, L. Wang, *Nat. Commun.* **2020**, *11*, 5437.
- [2] A. Amiri, R. Shahbazian-Yassar, *J. Mater. Chem. A* **2021**, *9*, 782–823.

- [3] K. Gu, D. Wang, C. Xie, T. Wang, G. Huang, Y. Liu, Y. Zou, L. Tao, S. Wang, *Angew. Chem. Int. Ed.* **2021**, *60*, 20253–20258; *Angew. Chem.* **2021**, *133*, 20415–20420.
- [4] A. Sarkar, L. Velasco, D. Wang, Q. Wang, G. Talasila, L. de Biasi, C. Kübel, T. Brezesinski, S. S. Bhattacharya, H. Hahn, B. Breitung, *Nat. Commun.* **2018**, *9*, 3400.
- [5] Y. Zheng, Y. Yi, M. Fan, H. Liu, X. Li, R. Zhang, M. Li, Z.-A. Qiao, *Energy Storage Mater.* **2019**, *23*, 678–683.
- [6] Q. Wang, A. Sarkar, D. Wang, L. Velasco, R. Azmi, S. S. Bhattacharya, T. Bergfeldt, A. Düvel, P. Heitjans, T. Brezesinski, H. Hahn, B. Breitung, *Energy Environ. Sci.* **2019**, *12*, 2433–2442.
- [7] Z. Du, C. Wu, Y. Chen, Z. Cao, R. Hu, Y. Zhang, J. Gu, Y. Cui, H. Chen, Y. Shi, J. Shang, B. Li, S. Yang, *Adv. Mater.* **2021**, *33*, 2101473.
- [8] K. Hurlbutt, S. Wheeler, I. Capone, M. Pasta, *Joule* **2018**, *2*, 1950–1960.
- [9] D. Su, M. Cortie, H. Fan, G. Wang, *Adv. Mater.* **2017**, *29*, 1700587.
- [10] M. Du, Q. Li, Y. Zhao, C.-S. Liu, H. Pang, *Coord. Chem. Rev.* **2020**, *416*, 213341.
- [11] G. Shen, Z. Liu, P. Liu, J. Duan, H. A. Younus, H. Deng, X. Wang, S. Zhang, *J. Mater. Chem. A* **2020**, *8*, 1154–1163.
- [12] J. Qian, C. Wu, Y. Cao, Z. Ma, Y. Huang, X. Ai, H. Yang, *Adv. Energy Mater.* **2018**, *8*, 1702619.
- [13] W. Jiang, T. Wang, H. Chen, X. Suo, J. Liang, W. Zhu, H. Li, S. Dai, *Nano Energy* **2021**, *79*, 105464.
- [14] H. Fei, J. Dong, Y. Feng, C. S. Allen, C. Wan, B. Voloskiy, M. Li, Z. Zhao, Y. Wang, H. Sun, P. An, W. Chen, Z. Guo, C. Lee, D. Chen, I. Shakir, M. Liu, T. Hu, Y. Li, A. I. Kirkland, X. Duan, Y. Huang, *Nat. Catal.* **2018**, *1*, 63–72.
- [15] Y. Pan, K. Sun, Y. Lin, X. Cao, Y. Cheng, S. Liu, L. Zeng, W.-C. Cheong, D. Zhao, K. Wu, Z. Liu, Y. Liu, D. Wang, Q. Peng, C. Chen, Y. Li, *Nano Energy* **2019**, *56*, 411–419.
- [16] H. Shang, X. Zhou, J. Dong, A. Li, X. Zhao, Q. Liu, Y. Lin, J. Pei, Z. Li, Z. Jiang, D. Zhou, L. Zheng, Y. Wang, J. Zhou, Z. Yang, R. Cao, R. Sarangi, T. Sun, X. Yang, X. Zheng, W. Yan, Z. Zhuang, J. Li, W. Chen, D. Wang, J. Zhang, Y. Li, *Nat. Commun.* **2020**, *11*, 3049.
- [17] H. Shang, W. Sun, R. Sui, J. Pei, L. Zheng, J. Dong, Z. Jiang, D. Zhou, Z. Zhuang, W. Chen, J. Zhang, D. Wang, Y. Li, *Nano Lett.* **2020**, *20*, 5443–5450.
- [18] Y. Zhu, W. Sun, J. Luo, W. Chen, T. Cao, L. Zheng, J. Dong, J. Zhang, M. Zhang, Y. Han, C. Chen, Q. Peng, D. Wang, Y. Li, *Nat. Commun.* **2018**, *9*, 3861.
- [19] A. Han, W. Chen, S. Zhang, M. Zhang, Y. Han, J. Zhang, S. Ji, L. Zheng, Y. Wang, L. Gu, C. Chen, Q. Peng, D. Wang, Y. Li, *Adv. Mater.* **2018**, *30*, 1706508.
- [20] Y. Ma, Y. Ma, S. L. Dreyer, Q. Wang, K. Wang, D. Goonetilleke, A. Omar, D. Mikhailova, H. Hahn, B. Breitung, T. Brezesinski, *Adv. Mater.* **2021**, *33*, 2101342.
- [21] W. Li, X. Guo, P. Geng, M. Du, Q. Jing, X. Chen, G. Zhang, H. Li, Q. Xu, P. Braunstein, H. Pang, *Adv. Mater.* **2021**, *33*, 2105163.
- [22] S. Zheng, Q. Li, H. Xue, H. Pang, Q. Xu, *Natl. Sci. Rev.* **2020**, *7*, 305–314.
- [23] Y. Bai, C. Liu, T. Chen, W. Li, S. Zheng, Y. Pi, Y. Luo, H. Pang, *Angew. Chem. Int. Ed.* **2021**, *60*, 25318–25322; *Angew. Chem.* **2021**, *133*, 25522–25526.
- [24] C. Guo, Y. Zheng, J. Ran, F. Xie, M. Jaroniec, S. Qiao, *Angew. Chem. Int. Ed.* **2017**, *56*, 8539–8543; *Angew. Chem.* **2017**, *129*, 8659–8663.
- [25] Y. Guo, J. Tang, Z. Wang, Y. Sugahara, Y. Yamauchi, *Small* **2018**, *14*, 1802442.
- [26] J. Nai, J. Zhang, X. W. David. Lou, *Chem* **2018**, *4*, 1967–1982.
- [27] Y. Duan, Z. Yu, S. Hu, X. Zheng, C. Zhang, H. Ding, B. Hu, Q. Fu, Z. Yu, X. Zheng, J. Zhu, M. Gao, S. Yu, *Angew. Chem.*

- Int. Ed.* **2019**, *58*, 15772–15777; *Angew. Chem.* **2019**, *131*, 15919–15924.
- [28] P. Geng, M. Du, C. Wu, T. Luo, Y. Zhang, H. Pang, *Inorg. Chem. Front.* **2022**, *9*, 2389–2394.
- [29] H. Jiang, X.-C. Liu, Y. Wu, Y. Shu, X. Gong, F.-S. Ke, H. Deng, *Angew. Chem. Int. Ed.* **2018**, *57*, 3916–3921; *Angew. Chem.* **2018**, *130*, 3980–3985.
- [30] J. Yu, J. Xiao, A. Li, Z. Yang, L. Zeng, Q. Zhang, Y. Zhu, L. Guo, *Angew. Chem. Int. Ed.* **2020**, *59*, 13071–13078; *Angew. Chem.* **2020**, *132*, 13171–13178.
- [31] T. Wang, Y. Li, J. Zhang, K. Yan, P. Jaumaux, J. Yang, C. Wang, D. Shanmukaraj, B. Sun, M. Armand, Y. Cui, G. Wang, *Nat. Commun.* **2020**, *11*, 5429.
- [32] J. Zhou, R. Li, X. Fan, Y. Chen, R. Han, W. Li, J. Zheng, B. Wang, X. Li, *Energy Environ. Sci.* **2014**, *7*, 2715.
- [33] T. Zhang, L. Zhang, L. Zhao, X. Huang, Y. Hou, *EnergyChem* **2020**, *2*, 100036.
- [34] J. Zhang, G. Li, Y. Zhang, W. Zhang, X. Wang, Y. Zhao, J. Li, Z. Chen, *Nano Energy* **2019**, *64*, 103905.
- [35] H.-G. Kim, H. Shin, Y. H. Ha, R. Kim, S.-K. Kwon, Y.-H. Kim, J.-J. Kim, *ACS Appl. Mater. Interfaces* **2019**, *11*, 26–30.
- [36] L. Ni, G. Zhao, Y. Wang, Z. Wu, W. Wang, Y. Liao, G. Yang, G. Diao, *Chem. Asian J.* **2017**, *12*, 3128–3134.

Manuscript received: June 26, 2022

Accepted manuscript online: August 25, 2022

Version of record online: September 8, 2022